

Epithelium Lab & Fundamentals of Histology/Pathology

Week 4 - SAL - Lecture 4 - faculty pathology deck + official Week 4 objectives

Master workflow: orient low power → classify architecture → inspect cells → relate to basement membrane/stroma → decide normal adaptation vs dysplasia/invasion.

Microscope + Digital Slide Rules

Microscope: start lowest objective, center/focus, then increase magnification. Fine focus at high power. Higher power = smaller field/depth.

Proscia: use minimap/thumbnail, zoom controls, annotation toggle/text. Start with architecture before cellular detail.

Specimen → Slide

Fixation (often buffered formalin) → gross exam/tissue selection → dehydration → clearing → paraffin infiltration → embedding → microtome sectioning (~4-10 µm in deck) → staining.

H&E + Special Stains

Tool	What it highlights	Color / clue
Hematoxylin	DNA/RNA, ribosome-rich structures	Blue-purple / basophilic
Eosin	Protein-rich cytoplasm, collagen	Pink-red / eosinophilic
PAS	Glycogen, neutral mucin, basement membrane, fungal walls	Magenta
PAS-D	Diastase removes glycogen	Compare before/after
Trichrome	Type I collagen	Fibrosis/connective tissue
Reticulin	Type III collagen	Black network
Elastic stain	Elastic lamina/fibers	Elastic architecture
Prussian blue / von Kossa	Iron / calcium	Pigment/mineral

Epithelium ID

Type	Recognition	Anchor
Simple squamous	1 layer, flat cells	Endothelium = vessels; mesothelium = serosa
Simple cuboidal	1 layer, round nuclei, height=width	Tubules/ducts; basement membrane at base
Simple columnar	1 layer, tall cells	Absorption/secretion; goblet cells possible
Pseudostratified	Nuclei at different heights; all cells touch BM	Often cilia + goblet cells
Stratified squamous	Multiple layers; surface cells flat	Abrasion protection

SAL Fast Recall

Basement Membrane

Anchors/supports epithelium, organizes polarity, separates it from connective tissue. Thin eosinophilic line may be seen; PAS can accentuate it; EM resolves ultrastructure. **Invasive carcinoma breaches it.**

Connective Tissue ID

Type	Pattern	Function
Loose/areolar	Loose fibers + ground substance	Flexible support, diffusion/immune traffic
Dense irregular	Collagen in many directions	Multidirectional stress
Dense regular	Parallel collagen	Unidirectional tensile strength
Elastic	Abundant elastic fibers/lamellae	Stretch + recoil
Adipose	Clear vacuolated cells; peripheral nuclei	Energy/cushion/endocrine

Pathology Language

Hypertrophy = ↑ cell size
Atrophy = ↓ cell size
Hyperplasia = ↑ cell number
Metaplasia = mature cell type replaced by another

Dysplasia = disordered atypical growth
Pleomorphism = variable size/shape
Anaplasia = loss of differentiation
High N:C = large nucleus, scant cytoplasm

Neoplasia / Malignancy

Term	Remember
Carcinoma	Malignant epithelial neoplasm
Adenocarcinoma	Malignant epithelial tumor with gland formation
Carcinoma in situ	Severe epithelial dysplasia without invasion
Invasion	Breaches normal confines/BM; may cause desmoplasia
Metastasis	Spread to distant/noncontiguous tissue

Case Anchors

Cervix: colposcopy + acetowhite change; LEEP excises lesion.
LSIL = low-grade; HSIL = high-grade squamous intraepithelial lesion.

Colon: adenomas are dysplastic/premalignant and often polypoid; colonoscopy can diagnose and remove them before invasion.

Must-Not-Miss

Simple vs stratified = layers. Pseudostratified still has every cell on BM. H&E = nuclei blue-purple / protein pink. Dense regular = parallel. Dysplasia ≠ invasion. Hypertrophy = bigger; hyperplasia = more. Start low power and think in 3D.

Source set: Week 4 official learning objectives + Barrett, Fundamentals of Pathology faculty deck.